

TRUPAL PATIL

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SUMMARY

Electrical engineer with hands-on equipment engineering experience in troubleshooting, tool qualification, and root cause resolution across high-volume manufacturing environments. Brings a background in cross-functional coordination, vendor management, and equipment reliability to support electrical equipment performance and process optimization.

EDUCATION

Arizona State University | MS – Management of Technology.

Tempe, AZ | Aug 2025 – May 2027

University of Mumbai | BE – Electrical Engineering

Mumbai, India | Jul 2016 – May 2019

EXPERIENCE

ASU Photovoltaic Reliability Lab | Research Aide

Mesa, AZ | Sep 2025 – Present

- Conducted structured reliability testing on PV devices, analyzing 70,000+ hourly data points across 8 performance parameters using Python and Excel to identify degradation trends and support equipment performance improvement.
- Refined analysis methodology and automated report generation using Python, reducing analysis time by ~5x while adapting priorities to support evolving research needs.

ReNew Pvt. Ltd | Sr. Equipment Engineer

Ahmedabad, India | Feb 2024 – Jun 2025

- Installed and qualified new cleanroom deposition tool sets across a 2.5GW fab ramp-up per planned schedule, ensuring on-time equipment readiness across all process modules.
- Reduced plant downtime from 8% to 3%, recovering ~438 production hours/year by handling on-duty alarm troubleshooting and resolving critical equipment issues through systematic root cause analysis.
- Cut repeat failure incidents by ~30% by building defense and monitoring systems to prevent equipment failure and performance loss, standardizing maintenance protocols across the fab.

Jupiter International Pvt. Ltd. | Sr. Equipment Engineer

Baddi, India | Jun 2023 – Feb 2024

- Achieved <2% process excursion rate (~27,000 cells/day within spec) from the first month by installing and qualifying low-pressure diffusion (DIF) tool sets per planned schedule.
- Reduced process excursions by ~15%, saving ~17,000 cells/day, by identifying hardware capability enhancements to optimize diffusion recipe parameters, temperature profiles, gas flow rates, and cycle times.
- Cut scheduled PM time by ~44 hours/tool/year (525 hours across the module) while maintaining 97%+ equipment availability by optimizing PM execution in coordination with facilities and manufacturing shift personnel.

PROJECTS

Reliability study of solar battery

- Evaluated lithium-ion battery reliability under thermal, humidity, and mechanical stress conditions through structured pre- and post-exposure characterization testing across multiple test cells.
- Delivered conclusions on battery degradation behavior and performance under real-world environmental stress, supporting reliability assessment for solar energy storage applications.

Solar shading of water canal

- Developed predictive equations modeling the relationship between solar shading and evaporation rate across sensor-collected data points over 13 months, informing water conservation strategy for surface water bodies in arid regions.

MOSFET Fabrication

- Built a strong understanding of semiconductor fabrication workflows through hands-on execution of the complete 20 nm MOSFET process flow, including integration of key cleanroom unit processes.

Strategic study of Lithography Industry

- Analyzed competitive technology roadmaps of leading lithography companies to understand how the global semiconductor industry sustains technological leadership.

SKILLS

Equipment Qualification (IQ/OQ/PQ) | Tool Installation & Ramp Support | Root Cause Analysis | Preventive Maintenance (PM) Program Development | Equipment Uptime & Reliability KPIs | Vendor & OEM Coordination | Process Excursion Troubleshooting | Tool-to-Tool Matching | Spare Parts Strategy | Mechanical/Electromechanical Systems | Cross-Functional Coordination (Process, Automation, Engineering) | Data Analysis (Python, SQL, Excel)